

Optimal UI Item Direction Definition (OUIID) in BiDi Interfaces: Method and Evaluation

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Abstract

The design process of BiDirectional (BiDi) interactive systems requires making decisions about the direction of the interface's UI items. The professional HCI literature traditionally assumes that each UI item has only one correct direction. We suggest that both directions can be appropriate for many UI items in BiDi interfaces, depending on contextual factors. This work introduces and evaluates the concept of optimal UI item direction (OUIID), a more nuanced, context-dependent perspective on the BiDi directionality problem. We evaluated the approach in an eye-tracking study with 30 UI items, by various measures of effectiveness, efficiency, and satisfaction. Using multi-objective optimization analysis across several measures, we demonstrate that OUIID varies depending on user expertise and prioritization of usability measures. The findings provide insights into design decisions in various setups.

CCS Concepts

• **Human-centered computing** → Human computer interaction (HCI); HCI theory, concepts and models.

Keywords

Bidirectional users, BiDi interface, UI element direction, Diversity, Usability, Optimization

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1 INTRODUCTION AND BACKGROUND

The global proliferation of interactive products, coupled with the growing emphasis on creating inclusive interfaces, has heightened the focus of developers and researchers on enhancing interface design for non-Western languages [5]. A notable group of such languages comprises bidirectional (BiDi) languages, including Arabic, Hebrew, Persian, and Urdu. These languages are primarily written from right-to-left (RTL) but occasionally require the incorporation of left-to-right (LTR) text elements, such as Latin script, numbers, and other notations, leading to mixed text directionalities, sometimes even within a single sentence [4]. Due to this unique structure, developing interfaces for interactive BiDi systems is associated with the challenge of determining the appropriate direction of certain UI items. Should the specific item be presented and read by RTL or LTR? The common approach to address this question assumes that the choice of UI item direction is dichotomic, and there is only one correct answer, either RTL or LTR. Research on the topic of BiDi design [10] is scarce, with most works adopting a perspective presuming the existence of a single correct direction [e.g., 19, 20]. For instance, one of the heuristics for Arabic mobile game design mentioned in [19] is “Aspire to RTL layout and navigation,” meaning that “An Arabic mobile game should be RTL justified in terms of its layout, images, scrolling, hierarchal menus, and navigation” [19, p.3].

In the absence of academic work on the topic, developers rely on BiDi guidelines from leading vendors to determine the UI direction in BiDi interfaces. However, these guidelines follow the dichotomic approach [2, 11, 18], mainly specifying item direction as either RTL or LTR. For instance, Google's Material Design guidelines state that “An illustrated sequence of events progresses RTL,” while charts and graphs should be presented in LTR [11].

Indeed, this approach is appropriate for some UI items for which recommendations are not controversial (e.g., when deciding on the overall layout direction of text content). Nevertheless, many UI items do not have a clear, unequivocal directionality. For such items, a case could be made for their presentation in either direction [7, 8], as demonstrated in Figure 1. Such elements typically contain dates (e.g., calendars), numbers (e.g., graphs), ranges (e.g., of numbers or dates), or images that are not directly related to the text (e.g., icons

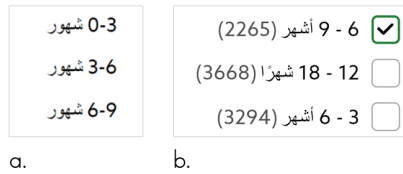


Figure 1: Example of two acceptable UI item directions from the Arabic version of NEXT.ae website: (a.) LTR direction for months' ranges on the home page; (b) RTL direction for months' ranges on the product page. In both, the Arabic labels, “months,” are in RTL. Retrieved July 25, 2024.

or navigation buttons). Goldenberg and Tractinsky [7, pp.12-13] present an extensive list of such UI items.

The multitude of UI items whose directionality is not clear-cut suggests a need to depart from the current approach to item directionality in BiDi interfaces, which assumes a single direction is correct. While the dichotomous approach remains prevalent in HCI literature, some studies acknowledge the possibility of dual-directionality and encourage designers to consider both options before making a choice. For instance, research on Arabic interface design patterns identified variations in chart directionality based on the data type represented on the X-axis [1]. Another example is a study on 16 directionality-ambiguous UI items conducted with 1705 Arabic and Hebrew users. The findings revealed that these items could be processed in either direction, with the interpretation influenced by factors such as overall content alignment, the type of UI element, age, and language [8].

The existence of UI items that can be presented and processed in both directions challenges the notion that items in BiDi interfaces require a single “correct” direction. Instead, we propose shifting focus to identifying the *optimal* direction. This perspective recognizes that both directions may be appropriate for many BiDi UI items but emphasizes considering *which option yields the best outcomes for specific users, contexts, and product requirements*. To address this, we introduce the concept of *Optimal UI Item Directionality* (OUIID). Following [15], we define OUIID as *the direction that best supports users in achieving their goals while considering effectiveness, efficiency, and satisfaction, given a specified context of use*.

In this work, we present a process for determining the OUIID and demonstrate its implementation, including identifying when directional ambiguity arise. Using an eye-tracking study with 42 BiDi-speaking users, we applied the OUIID concept to 30 UI items, evaluating them against nine usability measures across two use cases: novice users and users familiar with the task. We addressed the following research questions: 1. *Which of the two UI item directions is optimal for a novice user?* 2. *Which UI item direction is optimal for users familiar with the task?*

Our findings underscore the versatility of UI item direction determination in BiDi interfaces, demonstrating the benefits of adopting an optimality-based approach over the traditional dichotomous convention. This work contributes to the HCI community in the following ways: (1) We challenge the taken-for-granted dichotomic paradigm and introduce the OUIID approach as a more flexible and context-sensitive alternative. (2) We demonstrate the possibility of

flexible BiDi interface adaptation based on usability-related metrics and their priorities. At the same time, we emphasize the need to clearly define the metrics and their importance at the initial design stage. (3) Generally, our findings underscore the relationship between context and design and suggest a method to address design consistency.

2 DETERMINING OPTIMAL UI ITEM DIRECTION: PROPOSED APPROACH AND PROCESS

Following the context-dependent nature of HCI design, we propose that, in many cases, the direction is not an absolute and permanent trait of a BiDi UI item. Instead, it should often be evaluated case-by-case while considering various (potentially conflicting) criteria. Accordingly, one can assess the OUIID by allowing specific users to achieve their goals. We offer a four-step process to determine the OUIID, which can be repeated for different contexts, with different users, and criteria prioritization (see Figure 2).

The first step of the process is determining the crucial measures for the product or user goals. At this stage, one can use various objective and subjective measures such as error rate, user satisfaction, task completion time, etc. (see [13, 14]). In line with critical examinations of usability measures [14, 24], we posit that UI items that could be presented appropriately in either direction [7, 8] are likely to yield conflicting results on different measure criteria [12]. The explanation is simple: if all criteria favored one direction, that direction would dominate the other direction, and the item’s directionality would be indisputable. Hence, criteria prioritization is essential for determining the direction of ambiguous items (such as those listed in [7, 8]). The prioritization should be guided by product goals and use context (e.g., [9]). For instance, the design team could decide that reducing user errors is more critical than reducing task time. If the design concerns the directionality of multiple UI items, we recommend prioritizing them as well. For example, on a booking website, priority should be given to elements where misinterpreting directionality could lead to errors (e.g., date ranges), while lower priority can be assigned to accompanying graphical elements (e.g., departure and arrival icons).

In the second stage, RTL and LTR direction versions of the studied UI item are created and embedded in the UI. All other content elements (e.g., font, text alignment, boldness) remain constant and serve as the context for the evaluated item [7, 8]. In the third stage, we propose conducting a user study to evaluate the two variations of UI item direction.

The final stage is devoted to deciding the OUIID. At this stage, one can rely on data from statistical analyses. However, it is possible that conventional statistical methods may fail to reveal a significant difference between the RTL and LTR directions or that directional differences will not be consistent across measures (see, for example, [10]). To address this challenge, we propose adopting a Multi-objective optimization approach. Recent studies suggest the effectiveness of using various optimization approaches for UI design, targeting multiple measures [6, 16, 21]. This approach consolidates multiple measures into a single optimization score, weighted according to their relative importance [6], and allows for the prioritization of specific UI elements if needed. In cases where

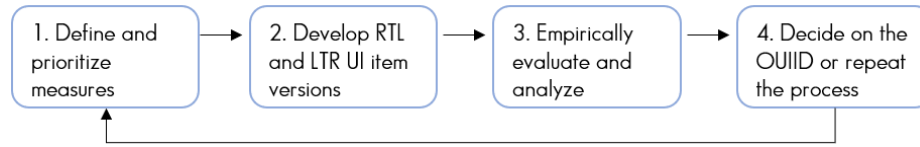


Figure 2: Proposed process for evaluating and determining the optimal UI item direction (OUID)

the optimization reveals no clear advantage for a specific direction, we propose incorporating consistency as a guiding factor [26] in the optimization process. Consistency is a critical requirement in interface design, influencing all dimensions of usability [17]. The optimization approach can balance consistency with other usability measures, maintaining directional consistency when no strong preference exists. Specifically, we suggest adhering to the following consistency hierarchy: first, maintain directional consistency within UI item sub-groups (see a list of sub-groups in [7, 8]) and then adhere to consistency with the direction of the remaining UI items. In the next section, we demonstrate a practical application of the OUID determination process.

3 DETERMINING THE OUID: EXAMPLE

To illustrate the process described in Section 2, we conducted a study to determine the optimal direction for a set of UI items in BiDi interface. Recognizing that prior experience and interface familiarity can influence BiDi users' preferences for item directionality [7], we applied the OUID approach under two conditions: one simulating novice users and the other simulating users familiar with the tasks.

Following the approach described in Section 2, we selected measures aligned with the three standard usability dimensions in the first stage. Effectiveness was assessed using error rate and the number of answer modifications. Efficiency was evaluated with task completion time and five eye-tracking metrics: fixations' number, number of fixations longer than 300 ms, fixation duration, blink rate, and pupil dilation. Satisfaction was examined through participants' feedback. Given the study's exploratory nature, we did not initially prioritize the measures.

In the second stage, we selected 30 UI items that, based on previous studies [7, 10], had two appropriate directions. We created RTL and LTR direction versions for each item, resulting in 30 stimulus pairs (see Appendix 1 for the complete list of UI items). Three of the items—*Backward and Forward*, *Hour Range String Ambiguous*, and *Scroll Bar*—were identified in previous research as having completely ambiguous directionality [7, 8], meaning they could be processed equally RTL or LTR. All stimuli were designed in black-white-blue colors with high contrast to meet accessibility requirements [25]. The only variation between the two item versions was direction; all other attributes, including label alignment, boldness, color, and content, were kept constant.

In the third stage of the OUID determination approach, we designed and conducted an eye-tracking experiment with 42 native BiDi-speaking students - 54.8% female, mean age 25.6 years (SD = 1.7, range 22-29). Detailed participant information, including age, specialization, color blindness, and use of corrective lenses, is

provided in the Supplementary Materials. The study was approved by the university's IRB for studies with human subjects.

The experiment was conducted in a windowless room with consistent lighting and air conditioning. We recorded eye movements using a desktop-mounted Gazepoint GP3 eye tracker (60 Hz) with the Open Gaze API. Stimuli were shown on a 20" LCD (1600×900 resolution) at a 65cm viewing distance, with participants instructed to minimize head movements to maintain calibration accuracy.

In the experiment, each UI item was paired with a task inspired by real-world usage examples. Appendix 1 provides a full list of tasks and stimuli used in the study. We created two study conditions, which were based on the expected optimal direction of the items. One condition presented items in their expected optimal direction (LTR or RTL), while the other presented them in the expected non-optimal direction (RTL or LTR, respectively). The expected optimal direction was determined using three sources in this order: findings from previous studies on BiDi users [7, 10], compatibility theory [22, 23], industry recommendations. We used our educated guess if no clear-cut direction emerged from the previous sources. Thus, each condition included both RTL and LTR items. Appendix 2 provides a list of the UI items and their expected optimal directions. The order of conditions was counterbalanced, with 52.4% viewing the optimal condition in the first trial and the non-optimal in the second trial. The other participants saw the conditions in reverse order. The trials were conducted one week apart. The first trial simulated novice users unfamiliar with the tasks and procedure; the second trial simulated experienced users.

The design of both trials was similar and included: (1) *Introduction*: following a 9-point eye-tracking calibration, the participants received an oral explanation about the experiment, completed a consent form and a vision evaluation, and answered demographic questions (orally). (2) *Eye-tracking experiment*: included a training phase with instructions and three practice tasks, followed by 30 main tasks in random order. The tasks involved reading a description, viewing a UI item (shown in LTR or RTL version according to the condition), and responding at their own pace by pressing the space bar. No time constraints were imposed, corrections were allowed, and participants could ask for clarifications. (3) *Post-Experiment Interview*: included two open questions: "Tell us about your experience?" and "Which tasks were the most difficult or unusual?" To control for the impact of fatigue on the findings, we asked participants to rate their fatigue orally on a 1–10 scale during the Introduction and Post-experiment interview for each trial. Independent t-tests revealed no significant differences in fatigue levels at the experiment's start between participants viewing optimal and non-optimal UI elements in the first trial.

Only valid eye-movement data, as identified by the eye-tracking system, were included in the analysis. One female participant (#24)

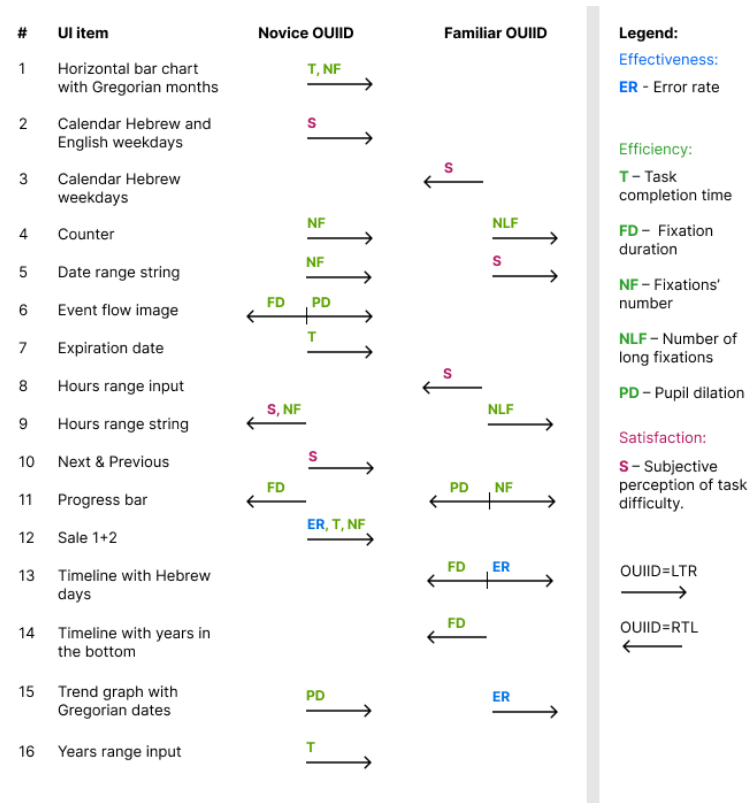


Figure 3: Summary of findings ($p < 0.05$) concerning differences between the versions of the UI item direction. Arrows represent the item's recommended direction. UI items and measures with no meaningful findings are omitted.

was excluded due to inaccuracies in gaze data recording. For the satisfaction measure quantitative analysis, we employed thematic analysis to group participants' comments regarding perceived task difficulty associated with the directional arrangement of each UI item. Subsequently, we quantified the number of negative statements for each UI item, condition, and use case (see Supplementary Materials for complete data). To compare the metrics expressed in different dimensions and eliminate bias due to individual differences, we normalized the quantitative data per participant using the percentile rank of each metric prior to the analysis to a scale between 0 and 1. The data was analyzed using a series of Mann-Whitney U tests for effectiveness and efficiency measures and a z-test for the satisfaction statements. Analysis was conducted for each UI item within each use case (trial). We used an alpha level of 0.05 without additional correction to identify meaningful differences, as each comparison was exploratory, independent, and not part of a holistic hypothesis [3]. Data analysis was performed using SPSS Statistics version 29.0. The independent variable was UI item direction version, and the dependent variables were the investigated measures. Meaningful differences ($p < 0.05$) for OUIID are summarized in Figure 3, the statistical data is presented in Appendix 3.

In the figure, arrows indicate items where significant differences were observed for a given use case and metric. We found no meaningful differences for 14 UI items and for the Answer modification

or Blink rate measures in either use case; therefore, these items and measures are not represented in the figure.

The statistical analysis revealed meaningful differences between direction versions for at least one measure in 13 out of 30 UI items for novice users, 8 out of 30 UI items for familiar users, and 16 out of 30 UI items overall. OUIID can be determined by the recommended direction given a specific measure. However, this becomes challenging when different measures suggest conflicting recommended directions. Indeed, we observed a *direction discrepancy between measures in 4 UI items* (#6, 9, 11, 13 in Figure 3). This highlights that both directions of these UI items are appropriate, with the ultimate selection of direction determined by the specific metric (and its priority) under consideration. It should be noted that *none of the UI items showed a meaningful difference in favor of one direction across all selected measures*. The lack of statistically significant results favoring a specific UI item direction reinforces the notion that no universally correct direction exists. This may erroneously lead to the conclusion that the direction of these UI items does not matter. However, this is not the case, as indicated in Section 1. Moreover, it might cause an inconsistent, problematic interface. Expanding the study sample may reveal more conclusive results; however, this alone is unlikely to provide clear guidelines for BiDi designers or serve as a feasible approach in practical settings. The nuanced nature of UI directionality highlights the importance of incorporating flexibility and context into design decisions.

To address the challenges of conflicting outcomes or lack of meaningful directional findings, we employed Multi-Objective Optimization [6] with a “sub-group direction consistency” guiding factor. To demonstrate the OUIID variability, we used three optimization models: (1) Equal weights for all measures, including the item’s directional consistency with its sub-group consistency; (2) Like the first scenario, but without sub-group consistency (all measures=1, Consistency=0); (3) A parsimonious model incorporating classical usability measures and sub-group consistency (i.e., Error rate, Task completion time, Satisfaction=1, Consistency=1, all other measures=0). Optimizations were run for both use cases (novice and familiar), resulting in six total optimizations. Figure 4 presents the OUIID results for each setting.

In the figure, a light blue cell indicates that the LTR direction was optimal, while a blue cell indicates that the RTL direction was optimal. The UI items in Figure 4 are grouped by sub-group and sorted by direction and optimality score: items with the highest scores for the LTR direction across all optimizations are placed at the top; items with the highest scores for the RTL direction are listed at the bottom.

The analysis still shows considerable variability but provides more precise indications of the UI directionality given the scenario (as reflected by the optimization model). We identified *six UI items with a consistent OUIID across both use cases and all models*. Specifically, the OUIID was consistently LTR for the Counter, Double slider, Expiration date input, Sale 1+2 string, and Trend graph with Gregorian weekdays. Conversely, the OUIID was consistently RTL for the Hours range input in all optimization scenarios.

OUIID variability was observed in 24 of the 30 UI items. For instance, in Toggle (#11 in Figure 4), the OUIID was LTR for novice users under Models 1 and 2 but switched to RTL under Model 3. For familiar users, the OUIID for Toggle was RTL under both Model 2 and 3 but LTR under Model 1.

For the four UI items where we observed discrepancies in OUIID between metrics in the statistical analysis (see 3), we also noted OUIID variability in the Multi-objective optimizations. In the *Progress bar* and *Hour range string* (#18 and #27 in Fig. 4), the OUIID varied between use cases and settings. In *Timeline Hebrew weekdays* and *Event Flow image* (#19 and #21 in Fig. 4), OUIID was RTL in five of six manipulations and switched to LTR in the familiar user case under the Model 3 setting. However, unlike the statistical analysis, Multi-objective optimization allowed us to conclude in favor of one direction while considering the context.

A noteworthy observation arose when comparing two unlabeled ranges (see Figure 5). While the *Hours’ range string* exhibited the expected RTL tendency (#28 in Fig. 4), the *Date range string* demonstrated an unexpected LTR tendency (#25 in Figure 4). For instance, speaking about LTR version of Date range string, one participant noted, “I know, it should be shown in Hebrew from RTL, but it is more intuitive for me to read the date range from LTR, I read in this direction automatically” (P3). Conversely, when the Hour range string was shown in LTR direction, the participants mentioned: “I expected the opposite direction” (P7), “It was the opposite of what I used to” (P39), and “I expected the right side to be the start and the left side to be the end” (P 20).

The difference between ranges was observed under Model 2 in both use cases and Model 3 in the familiar user use case. The use

of a forward slash and lack of spaces in the Date range likely led to its LTR processing (as a single LTR string), unlike the Hours’ range, which used a directionally neutral colon.

Another interesting finding is that in Model 3 (“classical measures” setting), in the novice user scenario, OUIID was predominantly RTL for 22 of 30 UI items (Figure 4, fifth and sixth columns). This finding suggests that the RTL version of these items is likely the most intuitive for users encountering them for the first time.

4 DISCUSSION AND LIMITATIONS

Our study challenges the traditional binary approach to UI item direction in BiDi interfaces by introducing and evaluating the concept of Optimal UI Item Direction. Our findings reveal that OUIID depends on the importance weights assigned to different measures. Therefore, we recommend prioritizing critical product measures early in the design process. This is particularly crucial for items for which the measures disagree on the optimal direction, such as Timeline Hebrew weekdays, Event flow image, and Progress bar (see Fig. 3, 4 for more such items).

Our findings, based on both statistical analysis and multi-objective optimization, revealed high variance in measurements, which sometimes allowed a single observation to overturn the OUIID decision. While this complicates drawing clear conclusions, designers still need to decide on directionality, even if the data cannot support them unequivocally. Including consistency as a guiding factor in the optimization model helped address this issue by favoring consistency within the same UI item group and prioritizing consistency over outliers. By introducing a consistency score, designers could prioritize uniformity in directionality, balancing the variance of results in specific metrics while optimizing the most critical items for overall user interaction. Consistency could be critical when there is a tradeoff between the recommended direction per item and holistic design consistency. Prioritizing a single item’s directionality may disrupt uniformity and confuse users. In turn, consistency may improve the measures across the study while observing the results across several items simultaneously and not independently.

Despite the differences in OUIID outcomes across various optimization models for most evaluated items, we observed complete consistency for six items across all multi-optimization models (see Figure 4). If this consistency is replicated in future studies, it would be appropriate to declare their direction as “correct.”

The study expands understanding of the OUIID problem and also contributes more evidence for the imperfect, context-dependent relationships between various usability measures [13, 14, 24]. We observed contradictions between metrics of different constructs and sometimes even within the same construct (Figure 3). Investigating these relationships across different settings can offer deeper insights and support more informed and effective UI design.

Our study has several limitations. (1) The study was conducted in a controlled lab environment, which may not fully reflect real-world distractions and conditions. At this stage, we consider it important to share the results of this study with the expectation that future research will relax some of the controls associated with laboratory studies to assess the findings’ generalizability to more real-world contexts. (2) We focused on individual UI item directionality but did

#	UI item	Novice User			Familiar user			OUIID tendency
		Model 1 (Equal weight)	Model 2 (Without consistency)	Model 3 (Classical measures)	Model 1 (Equal weight)	Model 2 (Without consistency)	Model 3 (Classical measures)	
BiDi text								
1	Expiration date							100% LTR
2	Sale 1+2							100% LTR
Performing mathematical tasks								
3	Counter							100% LTR
4	Double slider							100% LTR
Graphs LTR labels								
5	Trend graph with Gregorian dates							100% LTR
6	Horizontal bar chart with Gregorian months							83% LTR
Graphs RTL labels								
7	Horizontal bar chart with Hebrew months							83% LTR
8	Trend graph with Hebrew months							83% LTR
Navigation								
9	Backward & Forward buttons pair							67% LTR
10	Scroll bar (vertical)							67% LTR
11	Toggle							50% /50%
12	Next & Previous buttons pair							67% RTL
Order								
13	Numerical ranking							67% LTR
14	Star rating							50% /50%
15	Paginator							67% RTL
Time flow								
16	Event flow textually							50% /50%
17	Timeline - years in the top							67% RTL
18	Progress bar							67% RTL
19	Timeline - Hebrew days							83% RTL
20	Timeline - years in the bottom							83% RTL
21	Event flow image							83% RTL
Calendars								
22	Calendar Hebrew and English weekdays							67% RTL
23	Calendar without weekdays							67% RTL
24	Calendar Hebrew weekdays							83% RTL
Ranges								
25	Date range string							67% LTR
26	Date range input							67% RTL
27	Hours range string							67% RTL
28	Hours range ambiguous							83% RTL
29	Years range input							83% RTL
30	Hours range input							100% RTL

Legend:

OUIID =LTR

OUIID =RTL

Legend:
 OUIID =LTR
 OUIID =RTL

Figure 4: Results of six Multi-objective optimizations (three models each for two use cases – Novice and Familiar).



Figure 5: Two RTL range items, without labels, showed differences in OUIID. (a) Date range string: Hebrew label reads: “Your subscription is valid for.” (b) Hours range string: Hebrew label reads: “Emergency medical service working hours.”

not analyze OUIID at group or sub-group levels, which could reveal additional insights. (3) High variance in results sometimes impacted statistical conclusions, though incorporating a consistency factor helped address inconclusive cases. (4) The participants in the study consisted of a relatively homogeneous group of digitally literate students. Conducting similar studies with participants of other age groups, various BiDi languages, or lower digital literacy should help

validate our approach and findings, improve their generalizability, or identify their boundaries.

5 CONCLUSION

Common practice assumes that UI items in BiDi interfaces have a single correct direction—LTR or RTL. We challenged this perspective by arguing that many such items may be presented appropriately in either direction depending on context. We introduced OUID as a more context-sensitive approach to determining item directionality and demonstrated its sensitivity to context, use cases, measures, and prioritization. Our preliminary findings highlight the potential of integrating OUID and optimization methods into BiDi interface design practices and set the stage for benchmarking OUID in diverse contexts. Future product-specific experimentation may identify the optimal direction for its interface based on a blend of business and usability measures.

By presenting a nuanced, context-aware approach to BiDi UI item direction, we aim to foster discussions between researchers and industry practitioners. We hope such discussions will improve BiDi interface design and provide an opportunity to share insights on different usage scenarios and various BiDi languages. Overall, our findings suggest a broader and more flexible alternative to the dichotomous approach to UI item directionality in BiDi interfaces. Different usability measures across different items rarely align, making tradeoffs and prioritization necessary. As interface developers, we often seek a definitive answer to the question of the item direction. However, the answer most likely depends on the context, which must be defined before making a design choice. Then, by adhering to appropriate optimization principles and utilizing empirical data from previous studies, determining the optimal direction for each specific context becomes more informed and reliable.

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A. APPENDICES

A.1. Stimuli and tasks used in the Experimental part of the study

Table 1: Stimuli and tasks used in the Experimental part of the study

#	UI Item	Task	UI item direction version	
			LTR	RTL
Graphs LTR labels				
1	Horizontal bar chart with Gregorian years	A diagram of the number of new immigrants will be displayed on the next screen. According to the diagram, was the number of immigrants higher in 1949 than the following year?		
2	Trend graph with Gregorian dates	The next screen will show a chart representing a number of new coronavirus tests. According to the diagram, whether on 12.11 the number of verified people was higher than the number that was the previous day?		
Graphs Hebrew labels				
3	Horizontal bar with Hebrew months	A diagram of the number of publications in the faculty will be displayed on the next screen. According to the chart, was the number of publications in February higher than in January?		
4	Trend graph with Hebrew months	On the next screen, a chart will be displayed representing the number of ships arriving at the port. According to the chart, was the number of ships in May higher than the number of ships that arrived in April?		
Order				
5	Numerical ranking	On the next screen, you will see an option to rate the difficulty level of the test. Use the mouse to set the rating to 3.		
6	Paginator	On the next screen, a search results page will be displayed. You must return by clicking on the previous page.		
7	Star rating	On the next screen, you will see an option to rate the course you studied. Use the mouse to set the rating to 4 stars.		
Performing mathematical tasks				
8	Counter	An option to define the number of tickets will be displayed on the next screen. You need to press the button to increase the number of tickets to 2.		
9	Double Slider	A price range filter will be displayed on the next page. With the help of the mouse, show how to set the filter to the range between 0 and 60 NIS.		

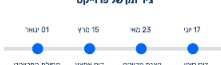
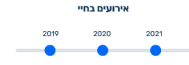
Ranges

10	Date range input	In the next screen the fields for entering a date range will be displayed. Let's say you were on vacation from October 23rd to October 30th. Click on the field where to enter your vacation start date (October 23rd)		
11	Date range string - only dates	An ad will be displayed on the next screen. You need to read it out loud.		
12	Hours' range input	The next screen will show an option to set delivery times. Suppose you need to set the time from 09:00 to 12:00. Click on the field where 09:00 should be written.		
13	Hours' range string ambiguous	An ad will be displayed on the next screen. You must read out loud.		
14	Hours' range string	On the next screen, information about the flight will be displayed. You need to say what time the landing will be.		
15	Years' range input	On the next screen, article filter fields will be displayed. Let's say you need to define a year range from 2000 to 2022. Click on the field where the last year in the range (2022) should be entered.		

BiDi t

16	Expiration date input	On the next screen, fields for entering credit card validity will be displayed. Click on the field where you need to type the year.		
17	Sale 1+2	An advertisement will be displayed on the next screen. You must read it out loud.		

Time flow

18	Event flow image	On the next screen, pictures of the daily actions of a girl will be shown. You have to look at the picture and say what action she did at first and what action she did afterward.		
19	Event flow textually	The next screen will show the steps in filling out a form. Look at the picture and say what details should be entered first.		
20	Progress bar	A progress bar will be displayed on the next screen. According to the bar, is the process closer to the beginning or the end?		
21	Timeline with Hebrew days	A project timeline will be displayed on the next screen. You need to say what the third phase of the project is.		
22	Timeline with years in the bottom	The next screen will show the order of events in life. Use the mouse to click on the first event.		
23	Timeline with years in the top	The next screen will show the order of events in life. Use the mouse to click on the last event.		

Calendars

24	Calendar Hebrew and English weekdays	A calendar will be displayed on the next screen. You need to look at the board and say what day of the week June 8, 2023, falls on.		
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25	Calendar Hebrew weekdays	A calendar will be displayed on the next screen. You must look at the board and say what day of the week February 22, 2023, falls on.		
26	Calendar without weekdays	A calendar will be displayed on the next screen. You must look at the board and say what day of the week March 15, 2022, falls on.		
Navigation				
27	Backward & Forward buttons pair (ambiguous)	A part of a website page will be displayed on the next screen. Click on the link that allows you to return to the previous page.		
28	Next & previous	The next screen will show a step in filling out a form. Click on the link that allows you to return to the previous step in the process.		
29	Scroll bar	A web page will be displayed on the next screen. You must click on the side where the scroll bar allowing the text to scroll down or up should be.		
30	Toggle	On the next screen, you will see an option to set notifications. You need to say what the active alert is.		

A2. UI item groups and sub-groups

Table 2: UI item groups and sub-groups

#	Group	Sub-group	UI item	Expected Optimal Direction	OUIID Tendency*		
1	Quantity, Order and Preference	Graphs LTR labels	•Horizontal bar chart Gregorian years	LTR	83% LTR		
			•Trend graph Gregorian dates	LTR	100% LTR		
			Graphs Hebrew labels	•Horizontal bar Hebrew months	RTL	83% LTR	
			•Trend graph Hebrew months	RTL	83% LTR		
		Order	•Numerical ranking	LTR	67% LTR		
			•Paginator	LTR	67% RTL		
			•Star rating	LTR	50%/50%		
		Performing mathematical tasks	•Counter	LTR	100% LTR		
			•Double Slider	LTR	100% LTR		
		2	BiDi text	Ranges	•Date range input	RTL	67% RTL
•Date range string	RTL				67% LTR		
•Hour range input	RTL				100% RTL		
•Hour range string ambiguous	RTL				83% RTL		
•Hour range string	RTL				67% RTL		
•Year range	RTL				83% RTL		
BiDi text	•Expiration date input				LTR	100% LTR	
•Sale 1+2	LTR			100% LTR			
3	Time flow			Time flow	•Event flow image	RTL	83% RTL
					•Event flow textually	RTL	50%/50%
		•Progress bar	RTL		67% RTL		
		•Timeline Hebrew days	RTL		83% RTL		
		•Timeline years in the bottom	RTL		83% RTL		
		•Timeline years in the top	RTL		67% RTL		
4	Calendars	Calendars	•Calendar Hebrew and English weekdays	LTR	67% RTL		
			•Calendar Hebrew	RTL	83% RTL		
			•Calendar without weekdays	LTR	67% RTL		
5	Navigation	Navigation	•Backward & Forward buttons	RTL	67% LTR		
			•Next & Previous buttons	RTL	67% RTL		
			•Scrollbar	LTR	67% LTR		
			•Toggle	RTL	50%/50%		

* Based on the results of Multi-objective optimization (See Section 4.4.2 for more information)

A3. Significant findings

Table 3: Significant findings (p <0.05) concerning differences between the versions of the UI item direction

#	UI item	Case	Effectiveness* ER	Efficiency*					Satisfaction**
				T	FD	NF	NLF	P	
1	Horizontal Bar chart Gregorian years	N	-	132.500, p=0.027	-	127.500, p=0.020	-	-	-
2	Calendar Hebrew and English weekdays	N	-	-	-	-	-	-	2.0997, p=0.04
3	Calendar Hebrew weekdays	N	-	-	-	-	-	-	2.3932, p=0.02
4	Counter	N	-	-	-	-	381.00, p=0.020	-	-
		F	-	-	-	134.500 p=0.049	-	-	-
5	Date range string - only dates	N	-	-	-	128.000 p=0.034	-	-	-
		F	-	-	-	-	-	-	2.0997, p=0.04
6	Event flow image	N	-	-	136.500, p=0.035	-	-	96.500, p=0.002	-
7	Expiration date input	F	-	121.500, p=0.022	-	-	-	-	-
8	Hour range input	N	-	-	-	-	-	-	2.0997, p=0.04
9	Hour range string	N	-	-	-	117.999 p=0.009	-	-	2.0997, p=0.04
		F	-	-	-	-	134.000, p=0.050	-	-
10	Next & previous	N	-	-	-	-	-	-	2.3128, p=0.02
11	Progress bar	N	-	-	115.000 p=0.008	-	-	-	-
		F	-	-	-	125.000, p=0.028	-	128.400, p=0.035	-
12	Sale 1+2	N	114.500, p=0.007	135.500, p= 0.033	-	138.500, p=0.040	-	-	-
13	Timeline Hebrew days	F	133.500, p=0.045	-	130.500, p=0.040	-	-	-	-
14	Timeline years in the bottom	F	-	-	83.00, p<0.001	-	-	-	-
15	Trend graph Gregorian dates	N	-	-	-	-	-	132.500 p=0.027	-
		F	113.500, p=0.011	-	-	-	-	-	-
16	Year range input	N	-	124.500 p=0.016	-	-	-	-	-

* Mann-Whitney U test, ** z-test. Abbreviations: ER – Error rate, T – Task completion time, FD – Fixation duration, NF – Fixations' number, NFL- Number of long fixations, P – Pupil dilation, S- Subjective perception of task difficulty; N- novice user case, F – familiar user use case.